HDOC DAY 56

PROBLEM

Write a C program that reads the size and elements of a variable-length integer array, arr[], and an integer k such that 0 < k ≤ size. Pass the array to a function named first_negative() using its address. Use pointers to pass the array between the calling function and first_negative(). The function first_negative() must:

- Consider every contiguous subarray of size k as a sliding window.
- Find the first negative integer in each window while moving from left to right.
- Display the first negative integer in the order of the corresponding windows.
- Display 0 if a window does not contain any negative integer.
- Separate the results by spaces.
- Display -1 if k is invalid.
- Use a nested-loop approach to solve the problem.

CODE

```
#include <stdio.h>


void first_negative(int *arr, int size, int k)
{
   if (k <= 0 || k > size)
   {
      printf("-1");
      return;
   }

   for (int i = 0; i <= size - k; i++)
   {
```

```
        int found = 0;

        for (int j = i; j < i + k; j++)
        {
            if (*(arr + j) < 0)
            {
                printf("%d ", *(arr + j));
                found = 1;
                break;
            }
        }

        if (found == 0)
        {
            printf("0 ");
        }
    }
}

int main()
{
    int size, k;

    printf("Enter the size of array: ");
    scanf("%d", &size);

    if (size <= 0)
    {
```

```
        printf("-1");
        return 0;
    }

    int arr[size];

    printf("Enter the array elements: ");
    for (int i = 0; i < size; i++)
    {
        scanf("%d", arr + i);
    }

    printf("Enter the window size k: ");
    scanf("%d", &k);

    first_negative(arr, size, k);

    return 0;
}
```